. A document that contains a 1) log of the testing you did (Prompts, outcomes (successful/unsuccessful). 2) Include the prompts you tested. If the agent fails, 3) log the prompt that that the agent failed on, 4) the error message if there is one, 4) what happened (e.g. asked a question requiring multiple regression and the agent suggested a bar plot; asked for an analysis for which there was a tool but the agent didn't find it and suggested code instead), and 5) where in the workflow you think it went wrong.

Command test

Help. Successful

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   Python

(.venv) (base) anhtrantt@Anhs-MacBook-Air QAC387-Group-Repo % OPENAI_API_KEY="sk-proj-5ZbLTHw3DpuxJOR50Z0SVnjh_YL-d62rREMIX32
KGk7ULYywfOfG451Ac_dmP7e7eoZzWy9YtgT3BlbkFJFu83zUsVMIQrwInq0saF0IcfxIUZsaaJZzvuQGc9xrHe0zdRDnlVj_mSeWJ4qk1xG2z9ROoegA" python
 builds/build3_hitl_router_agent.py --data data/sleep_mobile_stress_dataset_15000.csv --report_dir reports --tags stress_anal
ysis,build3 --memory
Langfuse: ENABLED (CallbackHandler + observe decorator)

Type 'help' for commands. Type 'exit' to quit.

> help

Commands:
  help                        Show this help text
  schema                      Print dataset schema
  suggest <question>          Build1-style suggestions (LLM)
  ask <request>               ROUTER decides: tool-run OR codegen (HITL)
  tool <request>              Force tool-run: choose one Build0 tool + args (HITL)
  code <request>              Force code generation (HITL) + approve to save
  run                         Execute last approved script via subprocess (HITL)
  exit                        Quit

Examples:
  ask run a frequency table for sex
  ask fit a regression of bill_length_mm on flipper_length_mm and sex
  tool run a correlation heatmap for numeric columns
  code create a plot of bill_length_mm by species and save it
```

Schema. Successful

```
PROBLEMS 3   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   Python

(.venv) (base) anhtrantt@Anhs-MacBook-Air QAC387-Group-Repo % OPENAI_API_KEY="sk-proj-5ZbLTHw3DpuxJOR50Z0SVnjh_YL-d62rREMIX32
KGk7ULYywfOfG451Ac_dmP7e7eoZzWy9YtgT3BlbkFJFu83zUsVMIQrwInq0saF0IcfxIUZsaaJZzvuQGc9xrHe0zdRDnlVj_mSeWJ4qk1xG2z9ROoegA" python
 builds/build3_hitl_router_agent.py --data data/sleep_mobile_stress_dataset_15000.csv --report_dir reports --tags stress_anal
ysis,build3 --memory


> schema

=== DATASET SCHEMA ===
Rows: 15000
Columns: 13

Columns and dtypes:
- user_id: int64
- age: int64
- gender: str
- occupation: str
- daily_screen_time_hours: float64
- phone_usage_before_sleep_minutes: int64
- sleep_duration_hours: float64
- sleep_quality_score: float64
- stress_level: float64
- caffeine_intake_cups: int64
- physical_activity_minutes: int64
- notifications_received_per_day: int64
- mental_fatigue_score: float64
```

Suggest <question>. “suggest factors leading to stress”. Successful

```
> suggest factors leading to stress
Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter
 or set LANGFUSE_PUBLIC_KEY environment variable.

### Research Questions:
- What is the relationship between daily screen time and stress levels among users?
  - **Outcome(s):** Stress level
  - **Predictor(s):** Daily screen time hours, phone usage before sleep minutes
  - **Suggested Analysis Type:** Correlation analysis or regression analysis

- How does physical activity influence stress levels in different age groups?
  - **Outcome(s):** Stress level
  - **Predictor(s):** Physical activity minutes, age
  - **Suggested Analysis Type:** ANOVA or interaction analysis

- Does caffeine intake correlate with stress levels and sleep quality?
  - **Outcome(s):** Stress level, sleep quality score
  - **Predictor(s):** Caffeine intake cups
  - **Suggested Analysis Type:** Multivariate regression analysis

### Clarifying Questions:
1. What specific aspects of stress are you interested in (e.g., general stress, work-related stress)?
2. Are there any particular demographic factors (like age or gender) you want to focus on in your analysis?
3. Do you have a specific hypothesis about how screen time or physical activity affects stress?
4. Are you interested in exploring the impact of sleep duration or quality on stress levels?
5. Would you like to include interactions between different predictors (e.g., age and physical activity) in your analysis?
6. Is there a particular time frame or context for the data collection that is relevant to your research?
7. Are you interested in visualizing the results, and if so, what type of visualizations do you prefer?
```

Ask “what do you think about this dataset”

- Did not use any summarize tools, instead suggested code

```
> ask what do you think about this dataset?

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Provide a basic overview of the dataset including its size, column names, and data types.",
  "note": "User is asking for a general overview of the dataset."
}

Note: User is asking for a general overview of the dataset.
```

  - But code worked correctly and did create a dataset overview csv in the reports folder.
- What happens if I prompt “ask summarize this dataset?”
  - Will being more explicit make it use summarise tools?

```
> ask summarise this dataset

=== ROUTER DECISION ===
{
  "mode": "tool",
  "tool": "basic_profile",
  "args": {},
  "note": "Basic profile provides an overview of the dataset."
}

Note: Basic profile provides an overview of the dataset.


=== TOOL PLAN (from router) ===
{
  "mode": "tool",
  "tool": "basic_profile",
  "args": {},
  "note": "Basic profile provides an overview of the dataset."
}

Note: Basic profile provides an overview of the dataset.

Run tool 'basic_profile' now? (y/n)
```

- Asking to summarize does lead to usage of a tool to summarise the dataset!
- So maybe it's about prompting better, and this isn't really an issue?

Ask predict my stress with 7 hours of sleep and 6 hours of screen time

```
> ask predict my stress with 7 hours of sleep and 6 hours of screen time

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Implement a linear regression model to predict stress level based on sleep duration and daily screen time. Use
the provided values of 7 hours of sleep and 6 hours of screen time for prediction.",
  "note": "No direct tool available for prediction; code generation is required."
}

Note: No direct tool available for prediction; code generation is required.


PLAN:
- Import necessary libraries (argparse, pandas, os, scikit-learn).
- Set up argument parsing for --data and --report_dir.
- Read the CSV file specified by --data using pandas.
- Check for missing values and handle them using listwise deletion.
- Validate that all required columns exist in the DataFrame.
- Implement a linear regression model to predict stress level based on sleep duration and daily screen time.
- Train the model using the relevant features.
- Use the model to predict the stress level for 7 hours of sleep and 6 hours of screen time.
- Save the prediction result as a CSV file in the specified --report_dir.
```

- Instead of using a new tool we made, predict_stress_from_sleep_phone, decided to generate code.
  - Maybe because I did not sat phone time and I said screen time?
  - So ask predict my stress with 7 hours of sleep and 6 hours of phone time

```
> ask predict my stress with 7 hours of sleep and 6 hours of screen time

=== ROUTER DECISION ===
{
  "mode": "codegen",
  "code_request": "Implement a linear regression model to predict stress level based on sleep duration and daily screen time. Use
the provided values of 7 hours of sleep and 6 hours of screen time for prediction.",
  "note": "No direct tool available for prediction; code generation is required."
}

Note: No direct tool available for prediction; code generation is required.


PLAN:
- Import necessary libraries (argparse, pandas, os, scikit-learn).
- Set up argument parsing for --data and --report_dir.
- Read the CSV file specified by --data using pandas.
- Check for missing values and handle them using listwise deletion.
- Validate that all required columns exist in the DataFrame.
- Implement a linear regression model to predict stress level based on sleep duration and daily screen time.
- Train the model using the relevant features.
- Use the model to predict the stress level for 7 hours of sleep and 6 hours of screen time.
- Save the prediction result as a CSV file in the specified --report_dir.
```

Ask What is the correlation between phone usage and sleep quality? Then code generation

```
> ask What is the correlation between phone usage and sleep quality?
Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter or
UBLIC_KEY environment variable.

=== ROUTER DECISION ===
{
  "mode": "tool",
  "tool": "pearson_correlation",
  "args": {
    "x": "phone_usage_before_sleep_minutes",
    "y": "sleep_quality_score"
  },
  "note": "Calculating the correlation between phone usage before sleep and sleep quality."
}

Note: Calculating the correlation between phone usage before sleep and sleep quality.

Router fallback: no matching tool is available in TOOLS. Falling back to code generation.

Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter or
UBLIC_KEY environment variable.

PLAN:
- Use argparse to handle command line arguments for --data and --report_dir.
- Read the CSV file specified in --data using pandas.
- Check for missing values in the relevant columns: 'phone_usage_before_sleep_minutes' and 'sleep_quality_score'.
- If missing values are present, apply listwise deletion to remove any rows with missing data.
- Validate that the columns 'phone_usage_before_sleep_minutes' and 'sleep_quality_score' exist in the DataFrame.
- Calculate the correlation between 'phone_usage_before_sleep_minutes' and 'sleep_quality_score'.
- Save the correlation result as a text file in the specified --report_dir.
```

-Agent asked to confirm before executing code. This was successful and correlation was generated to a file in the reports folder. Run command also successful

```
> run
Execute build3_generated_analysis.py now? (y/n) y

Running generated script...

Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter or set LANGFUSE_P
UBLIC_KEY environment variable.
Finished. Return code: 0
Saved execution log to: reports/run_log.txt
```

```
QAC387-Group-Repo
correlation_report.txt U
reports > correlation_report.txt
1   Correlation between phone usage before sleep and sleep quality score: -0.3352833523151466
2
```

TESTING NEW TOOLS ----------

tool predict_stress_from_sleep_phone {"sleep_hours": 5, "phone_mins": 90}
tool predict_sleep_quality {"phone_mins": 45, "sleep_hours": 6.5}

- KeyError: "Input to ChatPromptTemplate is missing variables {'allow_str', 'tool_arg_hints'}. Expected: ['allow_str', 'schema_text', 'tool_arg_hints', 'user_request'] Received: ['schema_text', 'user_request']\nNote: if you intended {allow_str} to be part of the string and not a variable, please escape it with double curly braces like: '{{allow_str}}'.\nFor troubleshooting, visit: https://docs.langchain.com/oss/python/langchain/errors/INVALID_PROMPT_INPUT "

- Fixed it by changing it so it no longer expected the two variables 'allow_str' and 'tool_arg_hints' so now it works correctly!

predict_stress_from_sleep_phone

```
> tool predict_sleep_quality {"phone_mins": 45, "sleep_hours": 6.5}

=== TOOL PLAN ===
{
  "tool": "predict_sleep_quality",
  "args": {
    "phone_mins": 45,
    "sleep_hours": 6.5
  },
  "note": "This tool is designed to predict sleep quality based on phone usage before sleep and sleep duration."
}

Note: This tool is designed to predict sleep quality based on phone usage before sleep and sleep duration.

Run tool 'predict_sleep_quality' now? (y/n) y
/Users/nirvaanmehta/Library/Mobile Documents/com~apple~CloudDocs/Pyhton/QAC387-Group-Repo/.venv/lib/python3.13/site-packages/sklearn/utils/validatio
n.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(

=== TOOL COMPLETE ===
Tool: predict_sleep_quality
Outputs saved to: reports/tool_outputs
Figures saved to: reports/tool_figures


Saved tool output to: reports/tool_outputs/predict_sleep_quality_output.txt


=== INTERPRETATION & SUMMARY ===

1) What we ran: We conducted a predictive analysis to estimate sleep quality based on the number of minutes spent on the phone and the total hours o
f sleep.

2) Key results:
- Predicted sleep quality score: [Insert predicted score here, if available]
- Relationship observed: Increased phone usage correlates with lower sleep quality.
```

predict_sleep_quality

```
> tool predict_stress_from_sleep_phone {"sleep_hours": 5, "phone_mins": 90}

=== TOOL PLAN ===
{
  "tool": "predict_stress_from_sleep_phone",
  "args": {
    "sleep_hours": 5,
    "phone_mins": 90
  },
  "note": "This tool is designed to predict stress levels based on sleep duration and phone usage before sleep."
}

Note: This tool is designed to predict stress levels based on sleep duration and phone usage before sleep.

Run tool 'predict_stress_from_sleep_phone' now? (y/n) y
/Users/nirvaanmehta/Library/Mobile Documents/com~apple~CloudDocs/Pyhton/QAC387-Group-Repo/.venv/lib/python3.13/site-packages/sklearn/utils/validatio
n.py:2691: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(

=== TOOL COMPLETE ===
Tool: predict_stress_from_sleep_phone
Outputs saved to: reports/tool_outputs
Figures saved to: reports/tool_figures


Saved tool output to: reports/tool_outputs/predict_stress_from_sleep_phone_output.txt


=== INTERPRETATION & SUMMARY ===

1) What we ran: We conducted a predictive analysis to estimate stress levels based on the input variables of sleep hours (5 hours) and phone usage (
90 minutes).

2) Key results:
- Predicted stress level: [Insert predicted stress level here, if available]
- Correlation between sleep hours and stress: [Insert correlation value here, if available]
- Correlation between phone usage and stress: [Insert correlation value here, if available]

3) Interpretation: Based on the analysis, the predicted stress level indicates how your current sleep and phone usage patterns may be affecting your
 overall stress. Generally, less sleep and increased phone usage can lead to higher stress levels.

4) Caveats/assumptions:
```

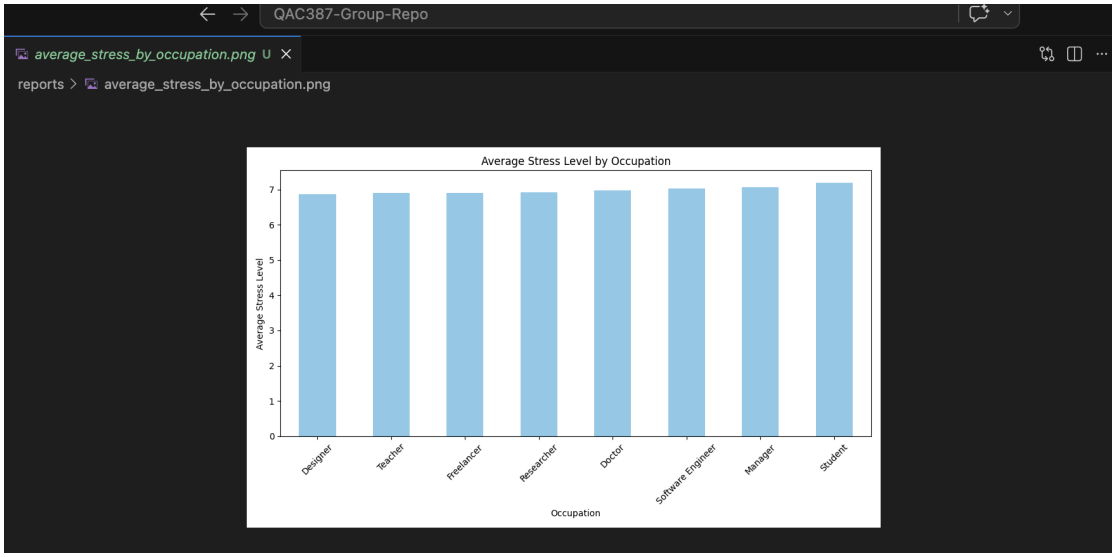
—CODE

code Create a bar chart of average stress by occupation

```
> code Create a bar chart of average stress by occupation
Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter or set LANGFUSE_P
UBLIC_KEY environment variable.

PLAN:
- Use argparse to handle command line arguments for --data and --report_dir.
- Read the CSV file specified in --data using pandas.
- Check for missing values in the 'occupation' and 'stress_level' columns.
- If missing values are present, apply listwise deletion to remove any rows with missing data.
- Validate that the columns 'occupation' and 'stress_level' exist in the DataFrame.
- Calculate the average stress level grouped by occupation.
- Create a bar chart of the average stress levels by occupation using matplotlib.
- Save the bar chart as an image file in the specified --report_dir.
```

Command successful and agent asked for verification before running code which outputted to correct folder

```
=== HUMAN VERIFICATION CHECKLIST (from model) ===
VERIFY:
- The script uses argparse to handle input arguments for the data file and report directory.
- It reads the specified CSV file using pandas.
- It explicitly checks for missing values and applies listwise deletion.
- It validates that the necessary columns exist before proceeding.
- The average stress levels by occupation are calculated and visualized in a bar chart.
- The bar chart is saved as an image file in the specified report directory.

Approve and save this code? (y/n) y

Approved and saved to: reports/build3_generated_analysis.py

Next: type 'run' to execute, or 'ask <request>' to route another request.

> run
Execute build3_generated_analysis.py now? (y/n) y

Running generated script...

Authentication error: Langfuse client initialized without public_key. Client will be disabled. Provide a public_key parameter or set LANGFUSE_P
UBLIC_KEY environment variable.
Finished. Return code: 0
Saved execution log to: reports/run_log.txt

>
```



Bar chart shows stress levels relatively unaffected by occupation, although designers have the lowest stress level and students the highest stress level

```
> exit
Goodbye!
(.venv) (base) anhtrantt@Anhs-MacBook-Air QAC387-Group-Repo %
```

Exit command functions as expected